

IS 3920 – Individual Project on Business Solutions
Software Requirements Specification
for
Smart Hospital Queue & Appointment Management
System

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[2026.01.17]

Student's Declaration

I hereby declare that this report is my original work and has not been submitted, in whole or in part, for any degree or diploma at any university or other institution of higher learning. All information derived from the work of others, whether published or unpublished, has been properly acknowledged in the text, and a complete list of references has been provided.

...2026.01.18.....
Date


.....
Signature of Student

Supervisors' declaration

I hereby declare that I have reviewed this project and find it adequate in both scope and quality.

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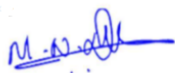
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Any further comments:

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Revision History

Name	Date	Reason for Changes	Version

1 Introduction

Healthcare institutions, particularly government hospitals in Sri Lanka, face significant challenges in managing patient flow efficiently. High patient volumes, limited resources, and manual administrative processes contribute to long waiting times, overcrowding, and dissatisfaction among patients and healthcare staff. Most government hospitals continue to rely on physical patient cards, manual appointment registers, and verbal calling systems, which are inefficient, error-prone, and lack transparency. These limitations negatively impact the overall quality of healthcare delivery and patient experience.

The **Smart Hospital Queue & Appointment Management System** is proposed as a digital solution to address these challenges by automating appointment handling, queue management, and patient tracking using modern web technologies. The system focuses on improving operational efficiency, reducing patient waiting times, and providing real-time visibility into hospital queue status through a centralized platform. This Software Requirements Specification (SRS) document formally defines the requirements of the proposed system.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete, precise, and unambiguous description of the functional and non-functional requirements of the Smart Hospital Queue & Appointment Management System. This document acts as a contractual reference between stakeholders, including developers, academic supervisors, hospital administrators, and evaluators, ensuring a shared understanding of system objectives and constraints.

The SRS serves multiple purposes throughout the system lifecycle. During development, it guides system design and implementation. During testing, it provides a benchmark against which system functionality can be validated. During maintenance, it assists future developers in understanding system behavior and design decisions. By clearly defining system requirements, this document minimizes ambiguity, reduces development risks, and supports successful project execution within the academic framework of IS 3920.

1.2 Document Conventions

This document adheres to the IEEE Software Requirements Specification (SRS) standard, ensuring consistency, clarity, and professionalism. The following conventions are used throughout the document:

- Functional requirements are stated using the keyword “shall” to indicate mandatory system behavior.
- Section and subsection numbering follows a hierarchical structure for ease of reference.
- Bullet points are used for listing objectives, features, and constraints.
- Technical terms and abbreviations are explained in Appendix A – Glossary.
- Diagrams referenced in the document are provided in Appendix B – Analysis Models and are designed to remain readable in printed format.

These conventions ensure that the document is easy to read, understand, and evaluate by both technical and non-technical stakeholders.

1.3 Intended Audience and Reading Suggestions

This SRS document is intended for a diverse audience involved in the development, evaluation, and usage of the system. The primary audiences include:

- **Project supervisors and mentors**, who assess the academic quality and feasibility of the project
- **Software developers and system designers**, responsible for implementing the system
- **Hospital administrators and IT staff**, who represent organizational requirements and constraints
- **Academic evaluators**, who assess compliance with IS 3920 project guidelines

Readers who are unfamiliar with the system domain are encouraged to begin with Section 1 (Introduction) to understand the context and motivation. Section 2 (Overall Description) provides a high-level overview of system architecture and users. Section 4 (System Features) should be referred to for detailed functional behavior, while Section 5 (Nonfunctional Requirements) outlines quality and performance expectations.

1.4 Product Scope

The Smart Hospital Queue & Appointment Management System is a web-based application designed to digitize and optimize patient queue and appointment handling in Sri Lankan government hospitals, with a primary focus on the National Eye Hospital, Colombo. The system aims to replace manual queue allocation, physical patient cards, and verbal calling mechanisms with a centralized digital solution that ensures transparency and efficiency.

The scope of the system includes online appointment booking, QR-based patient identification, real-time queue tracking, automated notifications, digital medical record access, and administrative reporting. The system does not aim to replace clinical decision-making processes but supports healthcare delivery by improving administrative efficiency and patient experience.

1.4.1 Problem in brief

Government hospitals in Sri Lanka experience severe congestion due to high patient demand and limited administrative automation. Patients often arrive early in the morning and wait for hours without clear information regarding their consultation time. Manual calling systems lead to confusion, missed turns, and inefficiencies. Physical patient cards are frequently lost or damaged, causing delays and incomplete medical history access for doctors.

Hospital staff face difficulties in managing queues fairly and efficiently, while doctors struggle with paper-based patient records that limit continuity of care. These challenges highlight the urgent need for a reliable, scalable, and digital solution that improves queue management and appointment handling.

1.4.2 Aim and Objectives

Aim:

To develop a Smart Hospital Queue & Appointment Management System that reduces waiting times and improves operational efficiency through web technologies and QR-based verification.

Objectives:

- To enable patients to book appointments online without visiting the hospital physically
- To generate unique QR codes for patient identification and arrival confirmation
- To provide real-time queue tracking with estimated consultation times
- To allow hospital staff to dynamically allocate and reschedule appointments
- To digitize patient medical records for easy access by doctors
- To replace manual calling systems with automated queue notifications
- To enhance transparency and accountability in government hospitals

1.5 References

- IEEE Software Requirements Specification Standard
- Ministry of Health Sri Lanka – Hospital Operations
- Related hospital queue management systems

2 Overall description

System architecture diagram

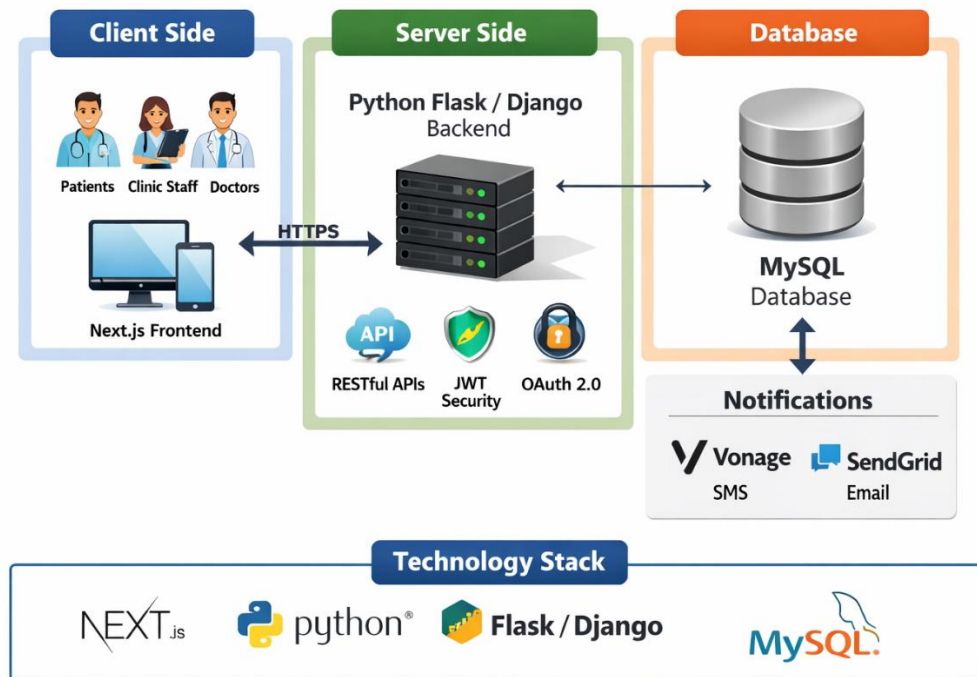


Figure 1 - System Architecture Diagram

The Smart Hospital Queue & Appointment Management System is designed using a Client–Server based architecture, which enables centralized data management, secure communication, and efficient handling of hospital operations. This architecture separates the user interface from backend processing, improving scalability, maintainability, and access control.

In this model, multiple user roles including patients, clinic staff, doctors, and administrators access the system through a web-based frontend, while a centralized server processes requests, applies business logic, and manages data storage. This approach is well suited for hospital environments where multiple users require real-time access to shared information.

The system architecture is organized into three main layers: Client Side (Frontend), Server Side (Backend), and Database with External Services. This clear separation of concerns enhances system reliability, supports future expansion, and ensures secure handling of sensitive medical data.

The frontend is developed using Next.js and provides responsive, role-based interfaces for all users. It communicates with the backend through secure RESTful APIs over HTTPS. The backend is implemented using Python frameworks such as Flask or Django and is responsible for business logic, authentication, authorization, and integration with notification services.

2.1 Product Perspective

The Smart Hospital Queue & Appointment Management System operates as a standalone, web-based information system that integrates into the existing administrative workflow of a hospital without disrupting clinical decision-making processes. The system does not replace medical diagnosis or treatment activities; instead, it supports healthcare delivery by digitizing appointment handling, queue management, and patient record accessibility.

The system follows a client–server architecture, which enables centralized data management and secure communication between users and the backend system. All user interactions occur through a web-based interface accessed via standard internet browsers. The backend server processes user requests, applies business logic, enforces security policies, and communicates with the database and external services.

From an architectural standpoint, the system is structured using a three-tier architecture, consisting of:

1. **Presentation Layer (Client Side)**
This layer includes all user-facing interfaces accessed through web browsers. It provides role-based dashboards for patients, clinic staff, doctors, and hospital administrators. The presentation layer is responsible for displaying information, capturing user inputs, and sending requests to the backend server.
2. **Application Layer (Server Side)**
The application layer contains the core business logic of the system. It handles user authentication and authorization, appointment scheduling, queue allocation, QR code generation, arrival verification, notification handling, and access to patient records. This layer ensures that all operations follow defined business rules and security policies.
3. **Data Layer (Database and External Services)**
The data layer manages persistent storage of patient details, appointment data, queue status, medical records, and system logs. It also integrates with external services such as SMS and email gateways to deliver notifications. Data integrity, confidentiality, and availability are ensured through controlled access and backup mechanisms.

This layered approach improves scalability, maintainability, and security, making the system suitable for high-volume hospital environments where multiple users access shared data concurrently.

2.2 Product Functions

The Smart Hospital Queue & Appointment Management System provides a set of integrated functions designed to automate and streamline hospital queue and appointment processes. Each function is carefully designed to reduce manual workload, minimize errors, and improve transparency for all stakeholders.

Appointment Booking and Registration

Patients can register on the system and book appointments online without physically visiting the hospital. This function reduces early-morning congestion and allows hospitals to plan patient flow in advance.

QR Code Generation and Validation

Upon successful appointment booking, the system generates a unique QR code for each patient. This QR code is used for patient identification, arrival confirmation, and queue activation, eliminating the need for physical patient cards.

Queue Number Assignment and Tracking

The system assigns queue numbers automatically based on appointment data and clinic schedules. Patients can view their queue position in real time, along with estimated consultation times.

Arrival Confirmation

Patients confirm their arrival at the hospital by scanning or submitting their QR code through the system. This ensures that only present patients are included in the active queue, reducing inefficiencies caused by absentee patients.

Real-Time Notifications

The system sends automated notifications via SMS and email to inform patients about appointment confirmations, queue status changes, doctor availability, and upcoming consultation turns.

Clinic Staff Queue Management

Clinic staff can manage queues using a dedicated dashboard that allows them to call patients, skip absent patients, reschedule appointments, and adjust queue flow dynamically.

Doctor Availability and Consultation Management

Doctors can update their availability, view patient lists, access medical histories, and mark consultations as completed. This improves workflow efficiency and continuity of care.

Digital Patient Record Management

The system stores patient medical records digitally, allowing doctors to access previous consultation details securely and quickly.

Administrative Monitoring and Reporting

Hospital administrators can monitor system usage, user activity, and operational performance. The system generates reports that support management decisions and policy planning.

2.3 User Classes and Characteristics

The system supports multiple user classes, each with distinct roles, responsibilities, and access privileges. Understanding these user classes is essential for designing appropriate interfaces and security controls.

2.3.1 Patient

Patients are individuals seeking medical consultation at the hospital. They may have varying levels of technical literacy; therefore, the patient interface must be simple, intuitive, and accessible. Patients primarily interact with the system to:

- Register and manage their profiles
- Book appointments online
- Receive and use QR codes
- Track queue status and estimated consultation time
- Receive notifications related to appointments

The system must ensure ease of use, minimal steps, and clear instructions for patients.

2.3.2 Clinic Staff

Clinic staff members are responsible for operational queue management within hospital clinics. They are expected to have basic computer literacy and familiarity with hospital administrative procedures. Their responsibilities include:

- Verifying patient arrivals using QR codes
- Managing live queues through dashboards
- Allocating and rescheduling appointment times
- Monitoring doctor availability

The system must provide clinic staff with efficient tools to manage high patient volumes accurately and fairly.

2.3.3 Doctor

Doctors are medical professionals who consult patients and require quick access to accurate patient information. Their interactions with the system include:

- Updating availability and consultation schedules
- Viewing patient lists and queue order
- Accessing digital medical histories
- Recording consultation outcomes

The system must prioritize reliability, data accuracy, and quick access for doctors to avoid workflow disruptions.

2.3.4 Hospital Administrator

Hospital administrators oversee overall system operation and user management. They have advanced system privileges and are responsible for:

- Managing user accounts and roles
- Monitoring system activity
- Generating operational and performance reports
- Ensuring policy compliance

Administrative interfaces must provide comprehensive visibility and control over system operations.

2.4 Operating Environment

- Web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge
- Frontend developed using Next.js(React framework), HTML5, and CSS3
- Backend developed using Python-based frameworks
- MySQL database server
- Internet-enabled devices including desktops, laptops, and tablet

2.5 Design and Implementation Constraints

The development and deployment of the system are subject to several constraints:

- **Data Privacy and Security:** Compliance with healthcare data confidentiality regulations
- **Infrastructure Limitations:** Limited IT resources in government hospitals
- **Academic Constraints:** Fixed project timeline and scope under IS 3920 guidelines
- **Third-Party Dependencies:** Dependence on external SMS and email services

These constraints influence design decisions and implementation strategies.

2.6 User Documentation

The system will provide:

- Online user manuals for each user role
- Step-by-step guides for common tasks
- Help and FAQ sections within the application

2.7 Assumptions and Dependencies

The system is developed based on the following assumptions:

- Users have access to internet-enabled devices
- Hospital management supports system adoption
- Users receive basic training where necessary

Key dependencies include:

- Availability of SMS and email gateway services
- Reliable internet connectivity within hospital premises

3 External Interface Requirements

This section describes all external interfaces of the Smart Hospital Queue & Appointment Management System. External interfaces define how the system interacts with users, hardware components, external software systems, and communication networks. Clearly specifying these interfaces ensures system interoperability, usability, security, and reliability within the hospital environment

3.1 User Interfaces

The system shall provide a web-based, responsive, and user-friendly interface designed to support multiple user roles, including patients, clinic staff, doctors, and hospital administrators. The user interface design prioritizes simplicity, accessibility, and clarity to accommodate users with varying levels of technical expertise.

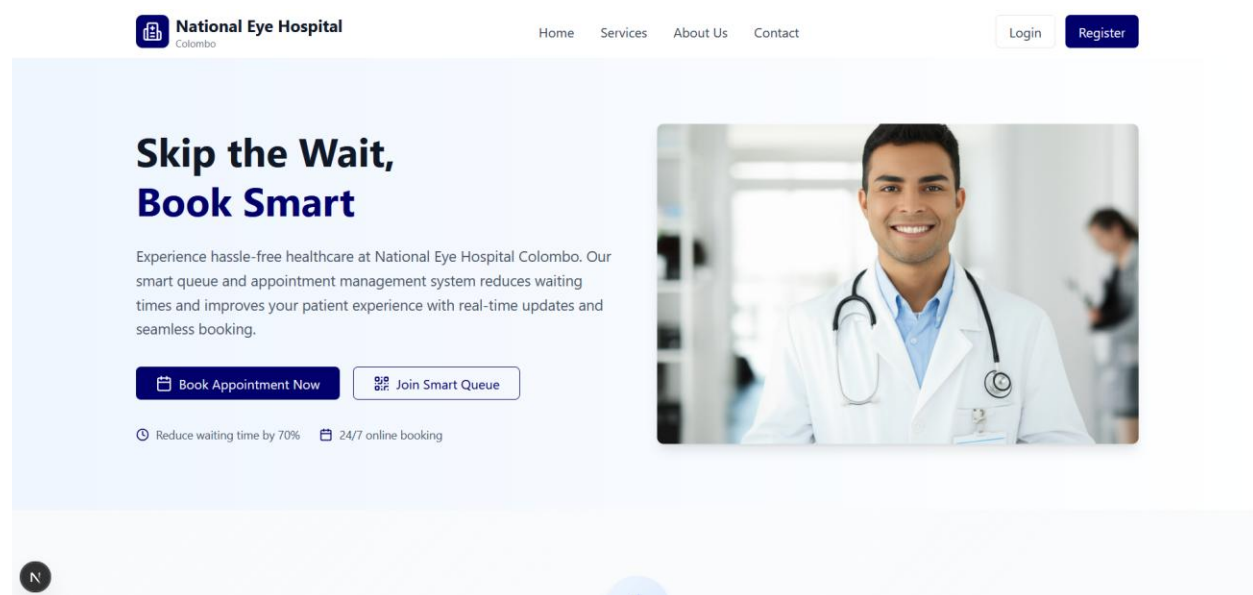


Figure 2- Home Page

← Back to Home

 **National Eye Hospital**
Colombo



Welcome Back

Sign in to access your patient portal

Login As

Patient

Doctor

Staff

Admin

Access your medical records and appointments

Patient ID, Email, or Phone

 Enter your patient ID, email, or phone

Password

 Enter your password 

☐ Remember me [Forgot password?](#)

Sign In to Patient Portal

 Patient Portal
Access your medical records and appointments

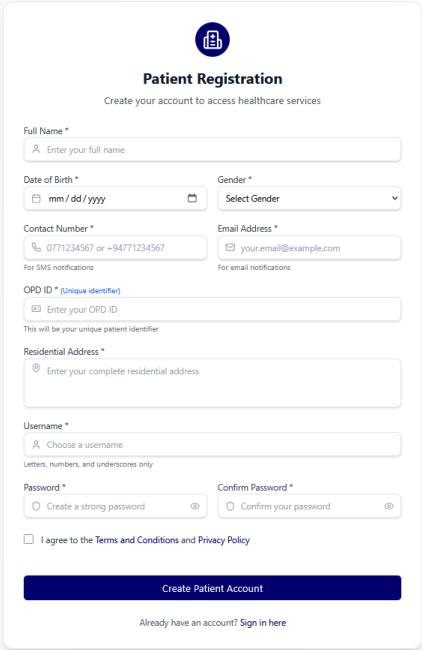
Figure 3 – Login Page

Patient Interface

The patient interface is designed to be minimalistic and intuitive, enabling ease of use even for first-time users. It shall allow patients to:

- Register and log in securely
- Book appointments online
- View assigned queue numbers
- Access QR codes for arrival confirmation
- Track real-time queue status and estimated consultation time
- Receive notifications related to appointments and queue updates

The interface shall use clear labels, readable fonts, and simple navigation to reduce user confusion and improve accessibility.



Patient Registration
Create your account to access healthcare services

Full Name *
Enter your full name

Date of Birth *
mm / dd / yyyy

Gender *
Select Gender

Contact Number *
0771234567 or +94771234567
For SMS notifications

Email Address *
your.email@example.com
For email notifications

OPD ID * (Unique Identifier)
Enter your OPD ID
This will be your unique patient identifier

Residential Address *
Enter your complete residential address

Username *
Choose a username
Letters, numbers, and underscores only

Password *
Create a strong password

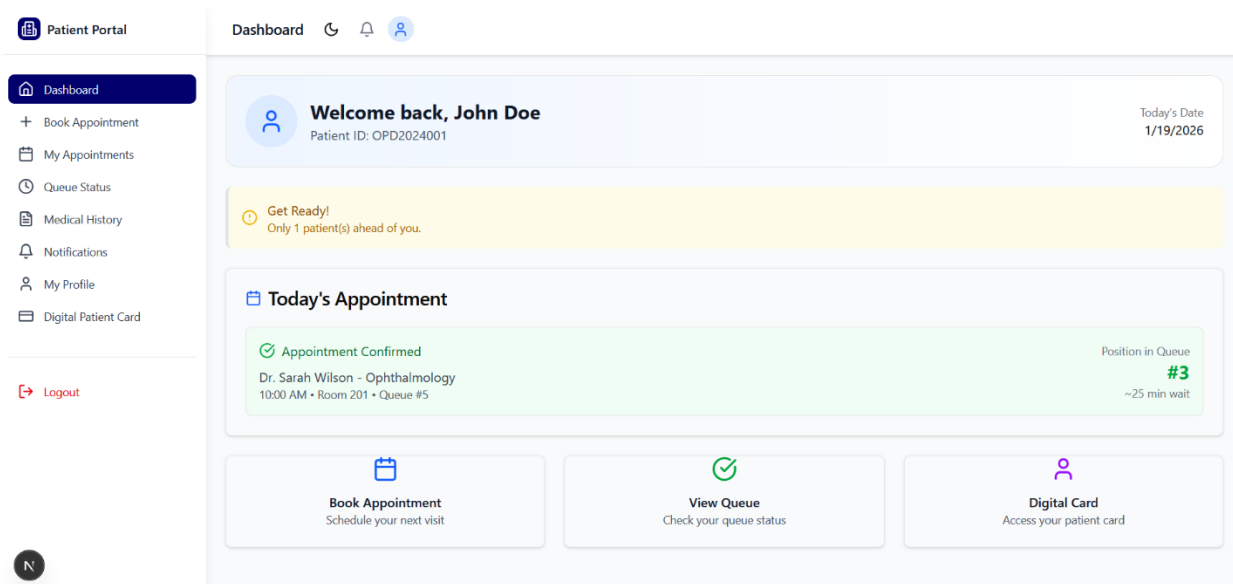
Confirm Password *
Confirm your password

☐ I agree to the [Terms and Conditions](#) and [Privacy Policy](#)

Create Patient Account

Already have an account? [Sign in here](#)

Figure 4 – Patient Registration Page



Patient Portal

Dashboard

- + Book Appointment
- My Appointments
- Queue Status
- Medical History
- Notifications
- My Profile
- Digital Patient Card

[Logout](#)

Welcome back, John Doe
Patient ID: OPD2024001

Today's Date
1/19/2026

Get Ready!
Only 1 patient(s) ahead of you.

Today's Appointment

Appointment Confirmed
Dr. Sarah Wilson - Ophthalmology
10:00 AM • Room 201 • Queue #5

Position in Queue
#3
~25 min wait

Book Appointment
Schedule your next visit

View Queue
Check your queue status

Digital Card
Access your patient card

Figure 5 – Patient Dashboard

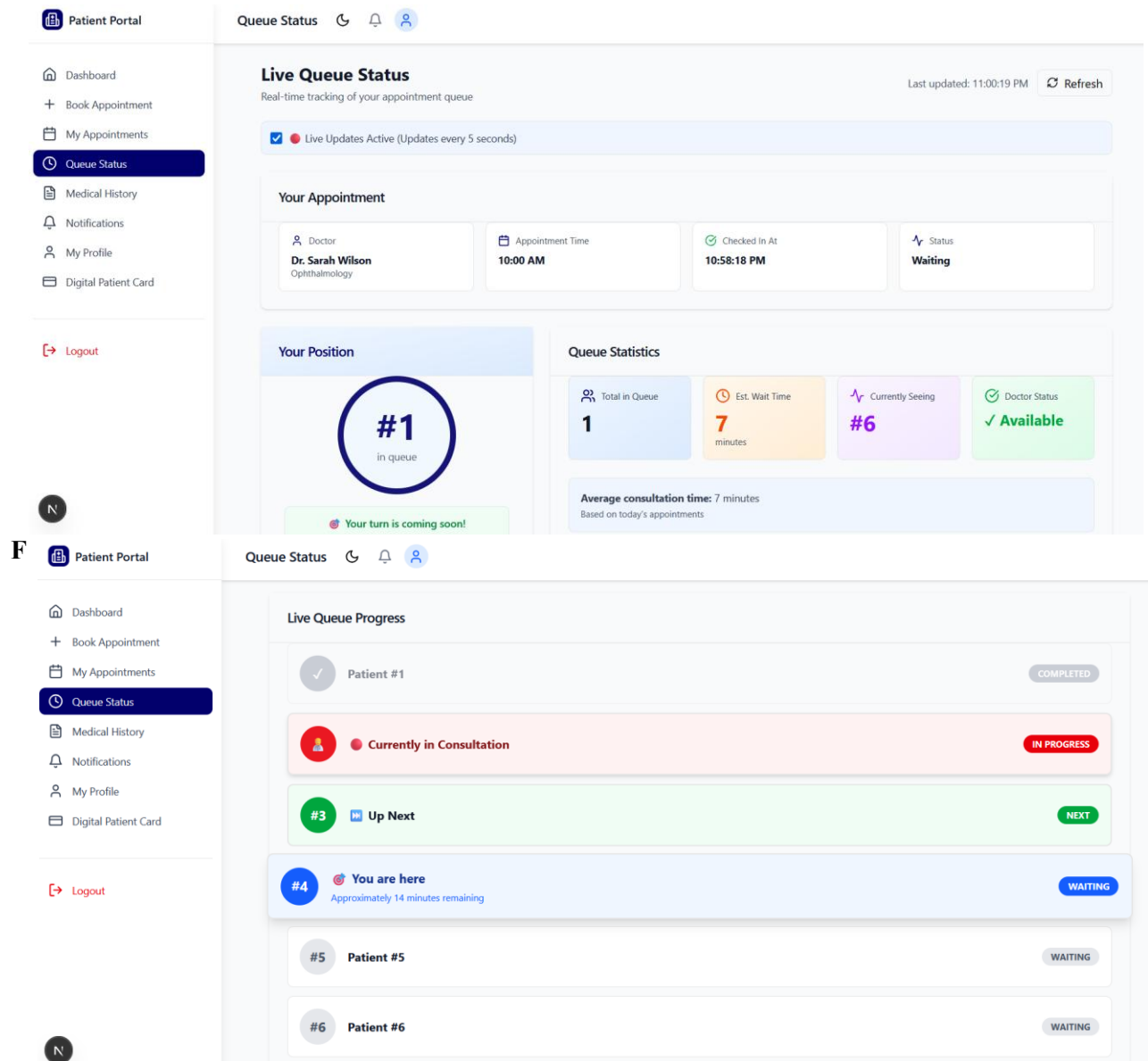


Figure 6 – Patient Live Queue

Clinic Staff Interface

The clinic staff interface shall provide a dashboard-focused layout with operational controls. It shall allow staff members to:

- Verify patient arrivals using QR codes
- View and manage live queues
- Call, skip, or reschedule patients
- Monitor doctor availability
- Update appointment allocations dynamically

This interface shall present real-time data and provide quick access to frequently used functions to support efficient queue management during high patient volumes.

Doctor Interface

The doctor interface shall be optimized for quick access to patient-related information. It shall allow doctors to:

- View daily patient lists and queue order
- Update availability status
- Access patient medical histories
- Record consultation outcomes
- Mark consultations as completed

The design shall minimize unnecessary navigation and ensure fast loading times to avoid disrupting clinical workflows.

Administrator Interface

The administrator interface shall provide comprehensive system oversight. It shall allow administrators to:

- Manage user accounts and roles
- Monitor system usage and activity logs
- Generate operational and performance reports
- Configure system settings

All user interfaces shall be responsive and accessible across desktops, laptops, and tablets.

3.2 Hardware Interfaces

The Smart Hospital Queue & Appointment Management System is designed to operate without mandatory specialized hardware. However, it may interface with commonly available hardware components to enhance usability and efficiency.

The supported hardware interfaces include:

- **QR Code Scanners:** Used by clinic staff to scan patient QR codes for arrival verification. These may include handheld scanners or mobile device cameras.
- **Computing Devices:** Desktop computers, laptops, and tablets used by staff, doctors, and administrators.
- **Mobile Devices:** Smartphones used by patients to access QR codes, notifications, and queue updates.

The system does not require direct interaction with medical devices or diagnostic equipment, ensuring ease of deployment in existing hospital environments

3.3 Software Interfaces

The system interacts with several internal and external software components to deliver its functionality. These software interfaces are defined to ensure interoperability and secure data exchange.

Backend APIs

The frontend communicates with the backend through RESTful APIs. These APIs handle:

- User authentication and authorization
- Appointment booking and queue management
- Patient data retrieval and storage
- Notification triggering

All APIs shall follow standard REST principles and return structured responses.

Authentication Services

The system shall use JSON Web Tokens (JWT) for secure user authentication and session management. JWT ensures stateless authentication and reduces server-side session overhead.

External Notification Services

The system integrates with third-party services to deliver notifications:

- **SMS Service:** Vonage API for sending real-time SMS alerts
- **Email Service:** SendGrid API for appointment confirmations and updates

These integrations ensure timely communication with patients and staff.

Database Interface

The system interfaces with a MySQL relational database for persistent data storage. The database stores patient profiles, appointment details, queue status, medical records, and system logs. Data access is strictly controlled through the backend to maintain integrity and security.

3.4 Communication Interfaces

All communication between system components and users shall be conducted over secure and reliable communication channels to protect sensitive healthcare data.

Network Communication

- The system shall operate over standard internet connections.
- All data transmission between clients and the server shall occur over **HTTPS**.
- Secure communication prevents unauthorized access and data interception.

Data Security and Encryption

- Sensitive data such as passwords and authentication tokens shall be encrypted.
- JWT tokens shall be securely generated and validated for each session.
- Access to APIs shall be restricted based on user roles.

Error Handling and Reliability

- The system shall handle communication failures gracefully.
- Users shall receive appropriate error messages in case of network issues.
- Retry mechanisms may be implemented for notification delivery failures

4 System Features.

This section describes the major functional features of the Smart Hospital Queue & Appointment Management System. Each system feature is explained in terms of its purpose, involved users, and functional requirements. These features collectively support the automation of appointment handling, queue management, and patient interaction while ensuring efficiency, transparency, and security.

4.1 User Authentication and Authorization

The User Authentication and Authorization feature ensures that only legitimate users can access the system and that each user can perform actions permitted by their role. This feature is essential for protecting sensitive patient data and maintaining system integrity.

The system supports role-based access control (RBAC) for four primary user roles: Patient, Clinic Staff, Doctor, and Hospital Administrator. Each role has clearly defined access privileges.

Actors

Patient

Clinic Staff

Doctor

Hospital Administrator

Functional Requirements

- The system shall allow users to register using valid personal and contact information.
- The system shall authenticate users using secure login credentials.
- The system shall generate and validate authentication tokens for active sessions.
- The system shall restrict access to system features based on user roles.
- The system shall allow users to log out and terminate active sessions.
- The system shall prevent unauthorized access attempts and log security events.

Priority High

4.2 Patient Registration and Profile Management

This feature allows patients to create and manage their personal profiles within the system. Patient profiles store essential information required for appointment booking, identification, and medical record association.

Digital patient profiles replace physical patient cards and reduce the risk of lost or damaged records.

Actors

Patient

Functional Requirements

- The system shall allow patients to register using personal details such as name, NIC/passport number, and contact information.
- The system shall allow patients to update profile details when necessary.
- The system shall store patient profiles securely in the database.
- The system shall associate medical records with the corresponding patient profile.
- The system shall prevent duplicate patient registrations.

Priority High

4.3 Appointment Booking and Queue Number Generation

This feature enables patients to book appointments online and receive queue numbers automatically. It reduces physical crowding at hospitals and allows staff to manage patient flow more efficiently.

Appointments are initially requested by patients and finalized based on doctor availability and clinic schedules.

Actors

Patient

Clinic Staff

Functional Requirements

- The system shall allow patients to request appointments through the web interface.
- The system shall validate appointment requests based on clinic availability.
- The system shall generate a unique queue number for each confirmed appointment.
- The system shall notify patients of successful appointment bookings.
- The system shall allow staff to reschedule appointments if required.

Priority High

4.4 QR Code Generation and Arrival Confirmation

This feature uses QR codes to confirm patient arrival and activate them in the live queue. QR-based verification replaces manual attendance marking and improves accuracy.

Each QR code is unique and linked to a specific appointment.

Actors

Patient

Clinic Staff

Functional Requirements

- The system shall generate a unique QR code for each confirmed appointment.
- The system shall allow patients to present QR codes upon arrival.
- The system shall allow staff to scan or verify QR codes.
- The system shall activate patients in the queue only after arrival confirmation.
- The system shall prevent duplicate or invalid QR code usage.

Priority High

4.5 Real-Time Queue Tracking and Notifications

This feature provides real-time visibility into queue status and estimated consultation times. Patients are informed about their position in the queue, reducing uncertainty and unnecessary waiting.

Automated notifications improve communication between the hospital and patients.

Actors

Patient

Clinic Staff

Functional Requirements

- The system shall display real-time queue positions to patients.
- The system shall calculate estimated consultation times dynamically.
- The system shall send notifications when a patient's turn is approaching.
- The system shall notify patients about doctor delays or schedule changes.
- The system shall support SMS and email notifications.

Priority High

4.6 Queue Management by Clinic Staff

Clinic staff manage live queues using a dashboard interface. This feature allows staff to handle real-world scenarios such as patient absence, emergency prioritization, and schedule adjustments.

Actors

Clinic Staff

Functional Requirements

- The system shall provide a live queue management dashboard.
- The system shall allow staff to call, skip, or reorder patients.
- The system shall allow staff to allocate or adjust appointment times.
- The system shall record queue handling actions for audit purposes.
- The system shall synchronize queue updates in real time.

Priority High

4.7 Doctor Availability and Consultation Management

This feature enables doctors to manage their availability and conduct consultations efficiently. Doctors can access patient details and update consultation status digitally.

Actors

Doctor

Functional Requirements

- The system shall allow doctors to update availability status.
- The system shall display patient lists based on queue order.
- The system shall allow doctors to access patient medical histories.
- The system shall allow doctors to record consultation outcomes.
- The system shall allow doctors to mark consultations as completed.

Priority High

4.8 Digital Medical Record Management

Digital medical record management ensures secure storage and retrieval of patient consultation data. It supports continuity of care and reduces dependence on paper-based records.

Actors

Doctor

Clinic Staff

Functional Requirements

- The system shall store medical records securely.
- The system shall associate records with patient profiles.
- The system shall restrict access to authorized users only.
- The system shall maintain a history of consultations.
- The system shall support future expansion for analytics.

Priority High

4.9 Administrative Monitoring and Reporting

Hospital administrators use this feature to monitor system usage, performance, and operational efficiency. Reports support decision-making and policy planning.

Actors

- Hospital Administrator

Functional Requirements

- The system shall provide dashboards for monitoring system activity.
- The system shall generate daily and monthly reports.
- The system shall log user actions and system events.
- The system shall support data export for analysis.

Priority Medium

5 Other Nonfunctional Requirements

Nonfunctional requirements define the quality attributes, constraints, and operational standards of the Smart Hospital Queue & Appointment Management System. While functional requirements describe what the system does, nonfunctional requirements specify how well the system performs its functions. These requirements are critical for ensuring reliability, security, usability, and long-term sustainability in a healthcare environment.

5.1 Performance Requirements

The system shall be designed to perform efficiently under normal and peak usage conditions, particularly in high-volume government hospitals where concurrent users are common.

- The system shall respond to user requests within three seconds under normal operating conditions.
- The system shall support multiple concurrent users, including patients, clinic staff, doctors, and administrators, without noticeable performance degradation.
- The system shall handle peak load conditions during clinic opening hours without system crashes or data loss.
- The system shall update queue positions and estimated consultation times in real time.
- The system shall ensure high availability during hospital operating hours.
- The system shall optimize database queries to reduce response latency.
- The system shall support future scalability to accommodate additional hospitals or clinics.

5.2 Safety Requirements

Although the system does not directly control medical equipment or clinical procedures, safety requirements are essential to prevent data loss, operational failures, and user confusion.

- The system shall maintain regular automated backups of all critical data, including patient records and appointment information.
- The system shall support data recovery mechanisms in the event of hardware or software failure.
- The system shall prevent inconsistent queue states caused by partial updates or system interruptions.
- The system shall display clear and informative error messages to users.
- The system shall ensure that critical operations such as queue updates and consultation completion are atomic and recoverable.
- The system shall maintain audit logs to support investigation in case of system failures.

5.3 Security Requirements

Security is a critical requirement due to the sensitive nature of healthcare data. The system shall implement strong security mechanisms to protect patient privacy and prevent unauthorized access.

- The system shall encrypt user passwords using secure hashing algorithms.
- The system shall use JWT-based authentication to manage user sessions securely.
- The system shall enforce role-based access control to restrict system functionalities.
- The system shall protect all data transmission using HTTPS encryption.
- The system shall prevent unauthorized access to patient medical records.
- The system shall log security-related events such as failed login attempts.
- The system shall automatically expire inactive user sessions.
- The system shall comply with applicable healthcare data privacy regulations and ethical standards.

5.4 Software Quality Attributes

The Smart Hospital Queue & Appointment Management System shall meet key software quality attributes to ensure long-term usability and maintainability.

Reliability

The system shall operate consistently without failure during hospital operating hours. Fault-tolerant mechanisms shall be implemented to minimize downtime.

Usability

The system shall provide intuitive interfaces that require minimal training. Clear navigation, readable layouts, and responsive design shall support users with varying technical skills.

Maintainability

The system shall be developed using modular architecture and clean coding practices, allowing future developers to easily understand, modify, and extend system functionality.

Scalability

The system shall support future expansion, including additional clinics, departments, or hospitals, without requiring major architectural changes.

Portability

The system shall be accessible across multiple platforms and browsers without requiring platform-specific configurations.

5.5 Business Rules

Business rules define organizational policies that the system must enforce to ensure correct operation and accountability.

- Only patients who have confirmed their arrival using a valid QR code shall be included in the active queue.
- Patients who fail to confirm arrival within a defined time window may be skipped or rescheduled.
- Doctors shall mark consultations as completed before closing appointments.
- Clinic staff shall manage queue changes according to hospital policies.
- Administrative approval shall be required for creating clinic staff and doctor accounts.
- System-generated reports shall be used for monitoring hospital performance and decision-making.

Appendix A – Glossary

This glossary defines key terms, abbreviations, and concepts used throughout the Software Requirements Specification (SRS) for the Smart Hospital Queue & Appointment Management System. The purpose of this glossary is to ensure clarity and consistency for all readers, including technical and non-technical stakeholders.

Administrator

A privileged system user responsible for managing user accounts, monitoring system activity, generating reports, and configuring system settings.

Appointment

A scheduled request by a patient to receive medical consultation at a specific clinic, managed digitally through the system.

Authentication

The process of verifying the identity of a user attempting to access the system, typically through login credentials.

Authorization

The mechanism that determines what actions an authenticated user is allowed to perform based on their assigned role.

Backend

The server-side component of the system responsible for business logic, data processing, authentication, authorization, and communication with the database and external services.

Clinic Staff

Hospital personnel responsible for operational activities such as verifying patient arrivals, managing queues, and allocating appointment times.

Client–Server Architecture

A system architecture model where client devices request services from a centralized server that processes requests and manages data.

Consultation

The interaction between a doctor and a patient during which medical assessment, diagnosis, or treatment is provided.

Dashboard

A user interface that displays real-time data and controls relevant to a specific user role, such as queue status or patient lists.

Database

A structured collection of data stored electronically, used to store patient information, appointments, queue status, and medical records.

Doctor

A medical professional who consults patients, updates availability, accesses medical records, and records consultation outcomes using the system.

Frontend

The client-side component of the system that users interact with through a web browser, built using web technologies such as HTML, CSS, and JavaScript.

HTTPS (Hypertext Transfer Protocol Secure)

A secure communication protocol used to encrypt data transmitted between the client and server.

JWT (JSON Web Token)

A compact, secure token format used for authentication and authorization in web applications.

Live Queue

The active list of patients who have confirmed their arrival and are waiting for consultation.

Medical Record

Digitally stored patient health information, including consultation history and treatment notes.

Notification

An automated message sent to users via SMS or email to inform them about appointment status, queue updates, or system alerts.

Patient

An individual seeking medical consultation who uses the system to book appointments, confirm arrival, and track queue status.

Queue Number

A system-generated identifier indicating a patient's position in the consultation order.

QR Code (Quick Response Code)

A machine-readable code generated by the system and used for patient identification and arrival confirmation.

Real-Time System

A system that processes data and updates outputs immediately or within a very short time delay.

RESTful API

An application programming interface that follows REST principles to enable communication between software components.

Role-Based Access Control (RBAC)

A security mechanism that restricts system access based on user roles.

Scalability

The ability of the system to handle increased workload or expansion without significant performance degradation.

Session

The period during which a user is authenticated and actively interacting with the system.

System Feature

A specific functionality provided by the system to support user requirements.

User Interface (UI)

The visual and interactive elements through which users interact with the system.

Web-Based Application

A software application accessed through a web browser over the internet.

Appendix B – Analysis Models

Use Case Diagram

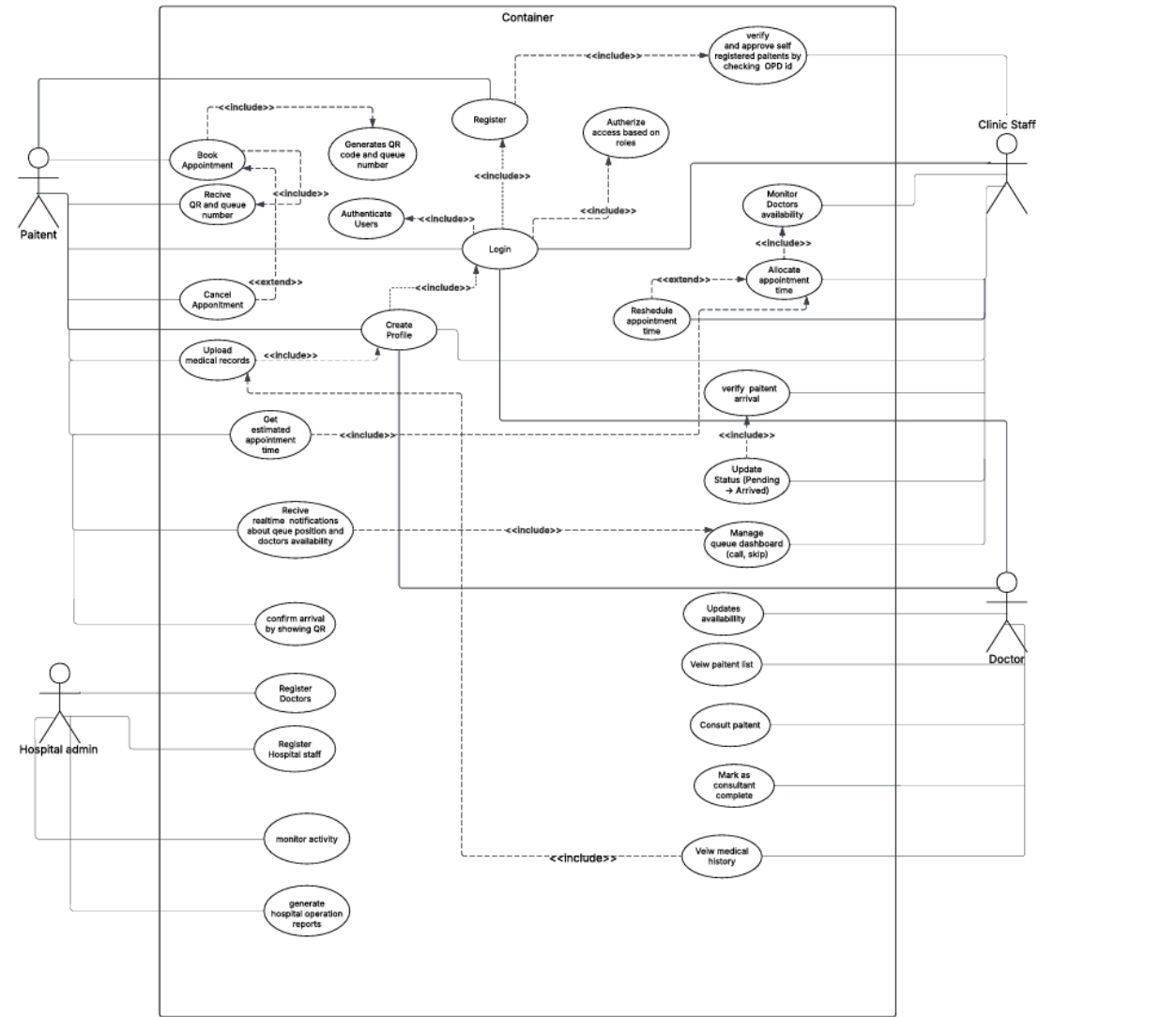


Figure 7 - Use Case Diagram

Link: [click here](#)

Class Diagram

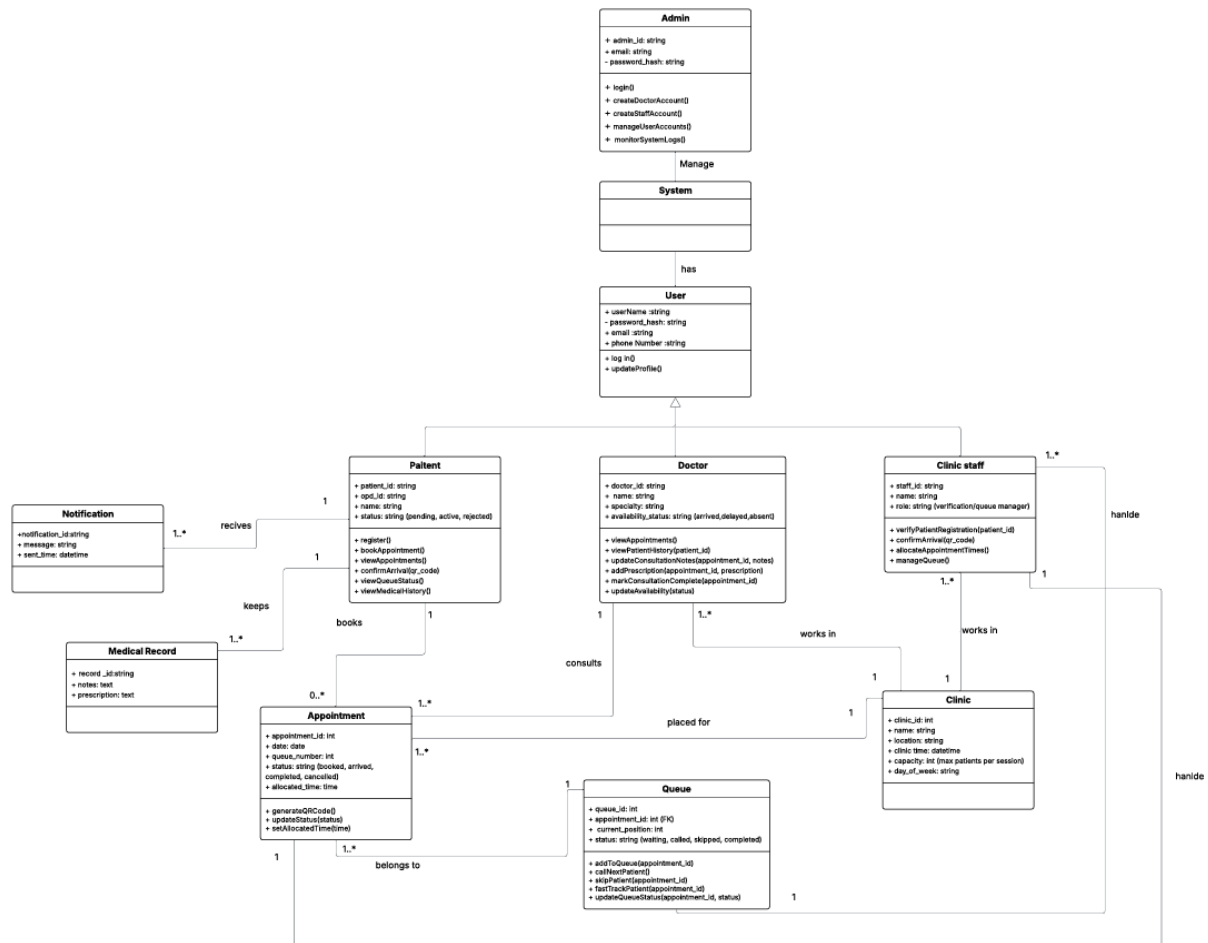


Figure 8 - Class Diagram

Link: [click here](#)

ER Diagram

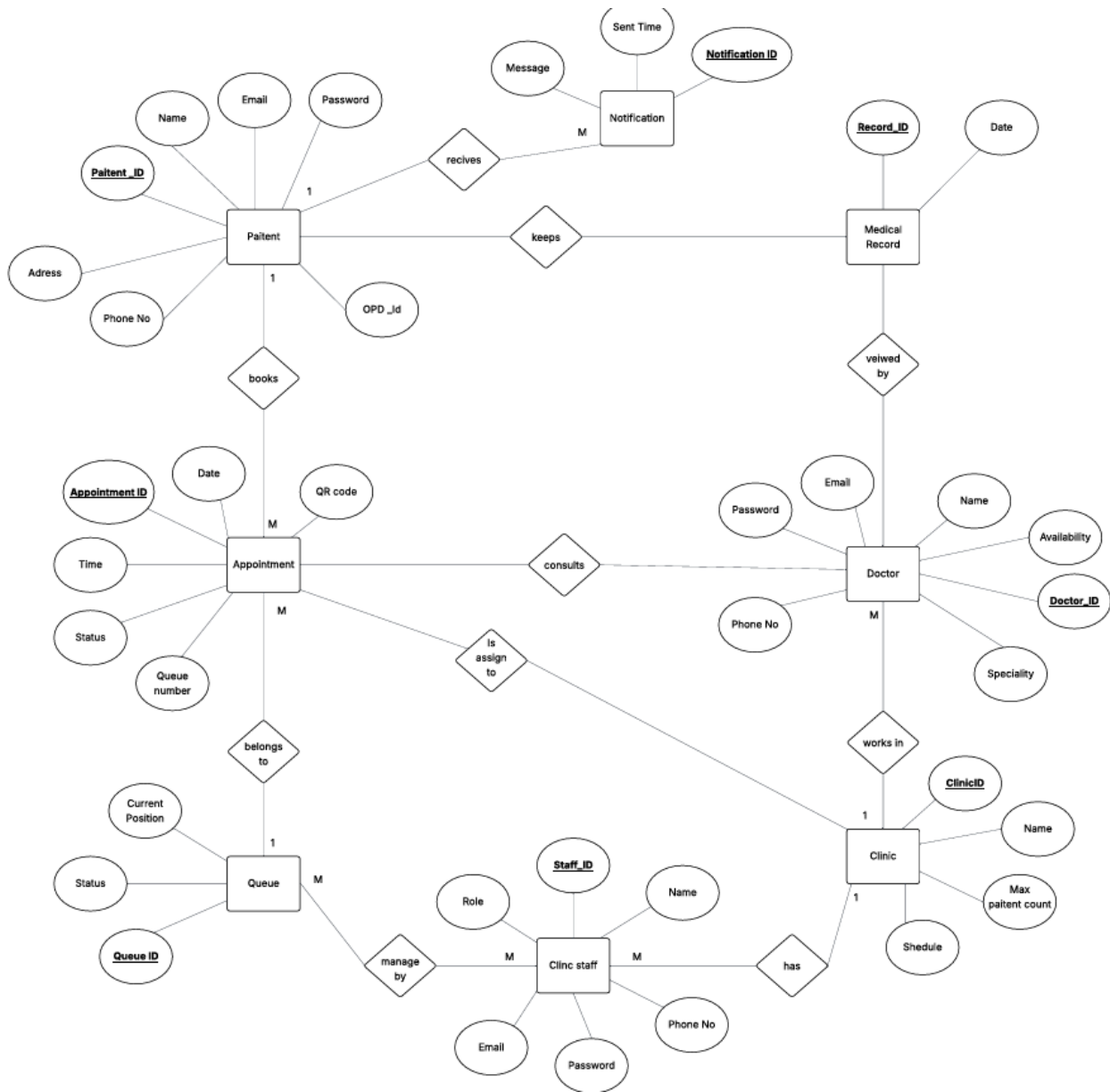


Figure 9 – ER Diagram

Link: [click here](#)